

Junjie (Takku) Li

@ alvinbluy@gmail.com | 📁 Portfolio | 🔗 LinkedIn | 🐙 GitHub | 📍 Seattle (Open to Relocate)

EDUCATION

Northeastern University

Master of Science in Computer Science

Teaching Assistant: Object Oriented Design, Algorithms

Sep 2023 - Aug 2025

Seattle, WA

Nankai University

B.S. in Mathematics

Tianjin, China

EXPERIENCE

Berkshire Hathaway HomeServices: Machine Learning Engineer Intern

Jun 2024 - Aug 2024

- Customized and fine-tuned **deep learning models** (N-Beats series) using Pytorch for housing price and inventory predictions, improving model accuracy by 30%. Applied **Agile** principles to iteratively refine data acquisition processes and update models, resulting in a 25% reduction in development cycle time.

Temple University: PhD Student in Quantitative Marketing

Sep 2021 - May 2023

Research contextual user engagement with real-time comments in movies.

- Explored the unique characteristics of real-time comments compared to after-consumption data (reviews, ratings) and to extract valuable insights such as emerging topics, emotions, and behavioral patterns.
- Applied a series of **BERT-based** models for content, sentiment, and emotion analysis, uncovering over 50% greater variance in contextual features compared to reviews.

PROJECTS

Movie Recommendation System | [Link](#)

Deep Learning Web

"VideoRecSys", a movie recommendation system with a personalized user experience.

- Designed a three-page frontend with a main page ranking movies by historical ratings, a detailed page showing similar movies via embedding-based content, and a "Guess You Like" page based on user interactions.
- Implemented key modules like feature engineering, recall, filter and re-rank for personalized contents. Powered dynamic adjustments using **Gradient Boosting decision trees** and **deep learning** models.
- Followed a robust deep learning architecture, integrating offline data processing on **Spark**, nearline stream processing with **Flink**, and online model serving using **TensorFlow Serving** and **MLeap**.

Search Engine for Emojis | [Link](#)

LLM-RAG Python/Go Web

"Emoji Search", a RAG-based AI search engine finding emojis based on web services.

- Adopted an event-driven architecture (**EDA**) for smoother search experience with four main components.
- Streamed user search queries through **Kafka** and integrated **Spark** to process large volumes of streaming data, improving real-time query handling by 20% and reducing batch processing time by 35% for analytics.
- Optimized microservice discovery and balanced queries, resulting in 30% fewer service failures and 15% faster response times at 5000 QPS. Containerized it in **Docker** and deployed it on **AWS EKS**.

AsyncTask Scheduler | Collaborated with [Hui Zhou](#) | [Link](#)

Distributed Go Framework

"Async4Ai", a light-weighted framework asynchronously processing interdependent tasks.

- Designed a three-page frontend featuring a main page with movie lists ranked by historical user ratings, a detailed page recommending similar movies using embedding-based content similarity, and a personalized "Guess You Like" page driven by user interactions and ratings.
- Crafted loosely coupled table normalization & sharding logic in **MySQL** to improve database performance.
- Performed stress testing and improved performance from 500 QPS to 2000 QPS by designing an efficient caching mechanism with **Redis**, using distributed locks and reconfiguring the database connection pool.

TECHNICAL SKILLS

Languages & Databases: Python(5yrs), Java, Golang, C++, JavaScript, SQL, Scala; MongoDB, Redis

Frameworks: CUDA, Dask, PyTorch, Flask, Spring Boot, Express, Node, React, Angular, FastAPI

Tools: Maven, JUnit, Postman, GitHub Actions, RabbitMQ, Kubernetes, Docker, Spark

Miscellaneous: Linux, AWS([Certified Solutions Architect Associate](#)), Azure ([Certified AI Engineer Associate](#))